

## ACCIDENTAL MECHANICAL SUFFOCATION IN INFANTS

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EACH year, for the past decade, an increasing number of children in the United States have been dying from accidental mechanical suffocation (Table I). It is of particular concern that of these fatalities from 70 to 80 per cent have occurred consistently in infants under 1 year of age. The literature has, however, paid but scant attention to accidental mechanical suffocation as a cause of death in early infancy.<sup>1</sup> The problem has remained, furthermore, a sadly neglected field in preventive medicine.

For example, consider the situation in the City of New York (Table II) where the annual deaths from accidental mechanical suffocation have exceeded those from measles, scarlet fever, and diphtheria combined and have equaled those from whooping cough.<sup>2</sup> Strikingly enough, here, too, it will be noted that about 80 per cent of the fatalities from accidental smothering involved infants under 1 year of age.

The problem, therefore, warrants more careful consideration than has been allotted to it in the past. For this reason, the case history records of 139 infants under 1 year of age who have died from accidental mechanical suffocation from January, 1939, to December, 1943, have been reviewed and are presented for discussion.\*† Of these fatalities 119 (85 per cent) occurred in white infants; 19 (15 per cent), in Negro infants; and a single case, in a Chinese baby. In 80 per cent of these babies both parents were native-born. In most instances the families were of the lower income groups.

### FACTORS CONTRIBUTING TOWARD ACCIDENTAL MECHANICAL SUFFOCATION

*Age and Sex.*—In Table III is shown the distribution of deaths according to age and sex. Eighty-five per cent of the fatalities occurred in infants under 6 months of age and 15 per cent, in those from 6 months to 1 year. Furthermore, the preponderance of deaths was in babies from 2 to 5 months (72 per cent) with the median age at death

This study was carried out under the direction of the Special Committee of Infant Mortality, Dr. Bret Ratner, Chairman, Medical Society of the County of New York.

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†The term accidental mechanical suffocation as used here refers to occlusion of the external openings (nose and mouth) to the air passages by some object so that smothering occurs by obstructive cutting off of air supply. Deaths from accidental mechanical suffocation by choking from solid objects or liquids lodging in the air passages are not included. (Gonzales, T. A., Vance, M., and Halpern, M.: *Legal Medicine and Toxicology*, New York, 1937, D. Appleton-Century Co., Inc., p. 274.)

TABLE I\*  
DEATHS FROM ACCIDENTAL MECHANICAL SUFFOCATION ACCORDING TO AGE AND SEX  
IN THE REGISTRATION STATES OF THE UNITED STATES (INCLUDING  
DISTRICT OF COLUMBIA), 1933 TO 1942

| YEAR | SEX   | ALL AGES | UNDER 1 YR. | 1 TO 4 YR. | 5 YR. AND OVER |
|------|-------|----------|-------------|------------|----------------|
| 1933 | Total | 934      | 692         | 27         | 215            |
|      | M     | 553      | 344         | 12         | 197            |
|      | F     | 381      | 348         | 15         | 18             |
| 1934 | Total | 1124     | 834         | 21         | 269            |
|      | M     | 704      | 451         | 13         | 240            |
|      | F     | 420      | 383         | 8          | 29             |
| 1935 | Total | 1230     | 875         | 27         | 328            |
|      | M     | 791      | 475         | 17         | 299            |
|      | F     | 439      | 400         | 10         | 29             |
| 1936 | Total | 1408     | 1007        | 36         | 365            |
|      | M     | 908      | 560         | 20         | 328            |
|      | F     | 500      | 447         | 16         | 37             |
| 1937 | Total | 1341     | 970         | 35         | 336            |
|      | M     | 885      | 574         | 13         | 298            |
|      | F     | 456      | 396         | 22         | 38             |
| 1938 | Total | 1269     | 1004        | 34         | 231            |
|      | M     | 780      | 565         | 18         | 197            |
|      | F     | 489      | 439         | 16         | 34             |
| 1939 | Total | 1226     | 1008        | 37         | 181            |
|      | M     | 744      | 570         | 23         | 151            |
|      | F     | 482      | 438         | 14         | 30             |
| 1940 | Total | 1489     | 1224        | 35         | 210            |
|      | M     | 876      | 685         | 16         | 175            |
|      | F     | 613      | 559         | 19         | 35             |
| 1941 | Total | 1523     | 1263        | 60         | 200            |
|      | M     | 933      | 721         | 42         | 170            |
|      | F     | 590      | 542         | 18         | 30             |
| 1942 | Total | 1581     | 1333        | 49         | 199            |
|      | M     | 977      | 777         | 29         | 171            |
|      | F     | 604      | 556         | 20         | 28             |

\*Compiled from Mortality Statistics and from Vital Statistics of the United States, U. S. Dept. of Commerce, U. S. Government Printing Office, Washington, D. C.

from 2 to 3 months. There was also a marked difference in deaths according to sex. Of the 139 fatalities, 69 per cent were in male babies and 31 per cent, in female babies. Thus, even in deaths from accidental mechanical suffocation the universal phenomenon of excess mortality in the male sex holds true.

*Season of Accident.*—The monthly variations in these deaths are shown in Fig. 1. It is not surprising that the peak of fatalities from accidental smothering took place during the winter months when more crib and carriage coverings are used to protect babies against the cold. This aspect of the problem will be discussed later.

*Hour of Accident.*—As will be readily understood, the exact hour of accidental suffocation could not be set with accuracy. In Fig. 2 is indicated the approximate time intervals at which the bodies were found after smothering took place. A sharp rise in deaths is evident between the hours from 3 A.M. to 6 A.M. and 6 A.M. to 9 A.M. This increase is worthy of note when considered in connection with the 2 A.M.

TABLE II\*

DEATHS FROM ACCIDENTAL MECHANICAL SUFFOCATION ACCORDING TO AGE AND SEX,  
NEW YORK CITY, 1934 TO 1943

| YEAR | SEX   | ALL AGES | UNDER 1 YR. | 1 TO 4 YR. | 5 YR. AND OVER |
|------|-------|----------|-------------|------------|----------------|
| 1934 | Total | 33       | 20          | 1          | 12             |
|      | M     | 19       | 11          | 0          | 8              |
|      | F     | 14       | 9           | 1          | 4              |
| 1935 | Total | 36       | 26          | 2          | 8              |
|      | M     | 22       | 14          | 2          | 6              |
|      | F     | 14       | 12          | 0          | 2              |
| 1936 | Total | 49       | 36          | 2          | 11             |
|      | M     | 31       | 20          | 1          | 10             |
|      | F     | 18       | 16          | 1          | 1              |
| 1937 | Total | 28       | 25          | 0          | 3              |
|      | M     | 21       | 18          | 0          | 3              |
|      | F     | 7        | 7           | 0          | 0              |
| 1938 | Total | 33       | 28          | 2          | 3              |
|      | M     | 17       | 14          | 0          | 3              |
|      | F     | 16       | 14          | 2          | 0              |
| 1939 | Total | 34       | 25          | 3          | 6              |
|      | M     | 24       | 19          | 1          | 4              |
|      | F     | 10       | 6           | 2          | 2              |
| 1940 | Total | 45       | 38          | 3          | 4              |
|      | M     | 26       | 21          | 1          | 4              |
|      | F     | 19       | 17          | 2          | 0              |
| 1941 | Total | 43       | 35          | 3          | 5              |
|      | M     | 25       | 20          | 3          | 2              |
|      | F     | 18       | 15          | 0          | 3              |
| 1942 | Total | 56       | 51          | 3          | 2              |
|      | M     | 37       | 33          | 3          | 1              |
|      | F     | 19       | 18          | 0          | 1              |
| 1943 | Total | 45       | 43          | 2          | 0              |
|      | M     | 33       | 32          | 1          | 0              |
|      | F     | 12       | 11          | 1          | 0              |

\*Compiled from Vital Statistics Tables, Bureau of Vital Records and Statistics, New York City Department of Health, Mr. Thomas J. Duffield, Registrar of Records.

and 6 A.M. feeding hours of the young infants and the times when adult members of the household are usually asleep.

*Vehicle of Suffocation.*—A study of the vehicles in which smothering occurred reveals that death took place mainly in cribs (35 per cent) and in the mothers' beds (15 per cent) during the morning hours (Table IV). In the afternoon and night hours the accidents ensued after placing the babies in carriages (23 per cent) and in cribs (19 per cent).

*Suffocating Agents.*—In Table V is listed the various agents found covering the heads of the babies or in occluding contact with the noses and mouths. In the crib, loose pillows, improperly applied blankets and mattress coverings (linen sheets and rubber sheetings), and mattresses were the common lethal agents. In the carriage, the unnecessary decorative pillows and coverings acted as the main agents in smothering. As has been previously noted, during the morning hours 15 per cent of the infants met death from overlaying after having been taken

for feeding into the mothers' beds. The smothering agent was either the mother's breast or upper extremity.

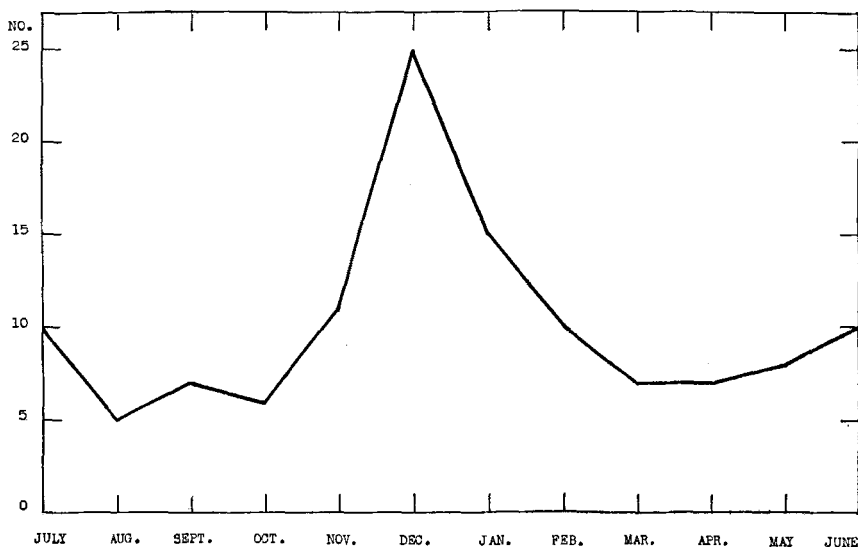


Fig. 1.—Accidental mechanical suffocation in infants under 1 year of age according to month of fatality.

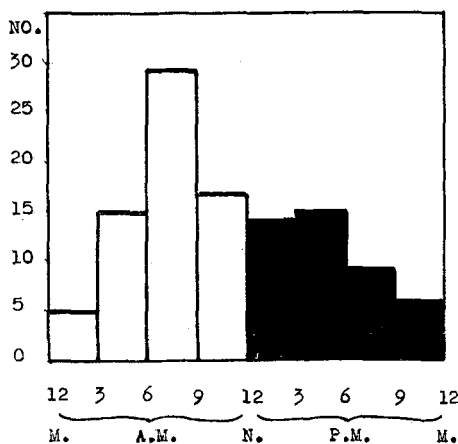


Fig. 2.—Time (hour) at which bodies of infants were found after occurrence of suffocation.

#### *Posture in Relation to Suffocation.—*

A. *The "Face-Down" Posture:* In Table VI are given the several postures in which the babies were found after accidental death. The "face-down" or prone posture was noted in 68 per cent of the fatalities. Of these babies 46 per cent were discovered with the noses and mouths in occluding contact with the lethal agents, usually soft pillows, mattresses, or mattress coverings. No additional agents covered the heads. In the remaining 22 per cent of the babies found "face-down," not only

TABLE III

DEATHS FROM ACCIDENTAL MECHANICAL SUFFOCATION IN 139 INFANTS UNDER 1 YEAR (ACCORDING TO AGE AND SEX)

| AGE (MO.) | SEX   | NUMBER | PER CENT |
|-----------|-------|--------|----------|
| 0 to 1    | Total | 14     | 10.0     |
|           | M     | 8      | 6.0      |
|           | F     | 6      | 4.0      |
| 1 to 2    | Total | 24     | 17.0     |
|           | M     | 14     | 10.0     |
|           | F     | 10     | 7.0      |
| 2 to 3    | Total | 34     | 25.0     |
|           | M     | 26     | 19.0     |
|           | F     | 8      | 6.0      |
| 3 to 4    | Total | 24     | 17.0     |
|           | M     | 17     | 12.0     |
|           | F     | 7      | 5.0      |
| 4 to 5    | Total | 18     | 13.0     |
|           | M     | 16     | 12.0     |
|           | F     | 2      | 1.0      |
| 5 to 6    | Total | 4      | 3.0      |
|           | M     | 1      | 1.0      |
|           | F     | 3      | 2.0      |
| 6 to 12   | Total | 21     | 15.0     |
|           | M     | 13     | 9.0      |
|           | F     | 8      | 6.0      |
|           | Total | 139    | 100.0    |
|           | M     | 95     | 69.0     |
|           | F     | 44     | 31.0     |

TABLE IV

VEHICLE OF SUFFOCATION IN 125 CASES OF ACCIDENTAL MECHANICAL SMOTHERING IN INFANTS UNDER 1 YEAR (ACCORDING TO TIME OF ACCIDENT)

| VEHICLE      | TIME (HR.) |          |        |          |
|--------------|------------|----------|--------|----------|
|              | A.M.       |          | P.M.   |          |
|              | NUMBER     | PER CENT | NUMBER | PER CENT |
| Crib         | 44         | 35.0     | 23     | 19.0     |
| Carriage     | 9          | 7.0      | 29     | 23.0     |
| Mother's bed | 19         | 15.0     | 1      | 1.0      |
| Total        | 72         | 57.0     | 53     | 43.0     |

TABLE V

AGENTS CAUSING SMOTHERING

| AGENTS                                              | VEHICLES |          |              |
|-----------------------------------------------------|----------|----------|--------------|
|                                                     | CRIB     | CARRIAGE | MOTHER'S BED |
| Blankets or coverings                               | 34       | 13       | 0            |
| Pillows                                             | 17       | 14       | 1            |
| Mattresses                                          | 20       | 6        | 0            |
| Mattress coverings (sheets, pads, rubber sheetings) | 12       | 6        | 0            |
| Mother (overlying)                                  | 0        | 0        | 19           |
| Sleeping bag                                        | 1        | 0        | 0            |
| Crib (wedging between sides)                        | 1        | 0        | 0            |
| Carriage (wedging between sides and hood)           | 0        | 1        | 0            |
| Bib                                                 | 1        | 0        | 0            |

TABLE VI  
POSTURES IN WHICH 128 BABIES WERE FOUND AFTER SUFFOCATION  
(ACCORDING TO VEHICLE OF DEATH)

|                                                                                           | VEHICLE      | NUMBER | PER CENT |
|-------------------------------------------------------------------------------------------|--------------|--------|----------|
| I. Postural Suffocation                                                                   |              |        |          |
| A. Face down or prone:                                                                    |              |        |          |
| 1. Nose and mouth in contact with suffocating agent                                       | Crib         | 44     | 35.0     |
|                                                                                           | Carriage     | 13     | 10.0     |
|                                                                                           | Mother's bed | 1      | 1.0      |
| 2. Nose and mouth in contact with one suffocating agent and head covered by another agent | Crib         | 17     | 13.0     |
|                                                                                           | Carriage     | 12     | 9.0      |
| B. Face up or recumbent:                                                                  |              |        |          |
| Face covered by lethal agent                                                              | Crib         | 14     | 11.0     |
|                                                                                           | Carriage     | 8      | 6.0      |
| II. Overlaying                                                                            | Mother's bed | 19     | 15.0     |

were the noses and mouths in occluding contact with the instruments of death but the heads were also covered by accessory agents, such as pillows or blankets. In passing, it should be pointed out that the predominant number of infants found in the prone posture fell into the age group below 5 months (Table VII).

TABLE VII  
POSTURE AT SUFFOCATION IN RELATION TO AGE AT DEATH

| AGE<br>(MO.) | FACE DOWN |          | FACE UP |          | MOTHER'S BED<br>(OVERLAYING) |          |
|--------------|-----------|----------|---------|----------|------------------------------|----------|
|              | NO.       | PER CENT | NO.     | PER CENT | NO.                          | PER CENT |
| 0 to 1       | 5         | 4.0      | 3       | 2.0      | 5                            | 4.0      |
| 1 to 2       | 11        | 9.0      | 5       | 4.0      | 7                            | 6.0      |
| 2 to 3       | 26        | 20.0     | 3       | 2.0      | 4                            | 3.0      |
| 3 to 4       | 16        | 13.0     | 5       | 4.0      | 3                            | 2.0      |
| 4 to 5       | 12        | 9.0      | 1       | 1.0      | 0                            | 0.0      |
| 5 to 6       | 4         | 3.0      | 0       | 0.0      | 0                            | 0.0      |
| 6 to 12      | 13        | 10.0     | 5       | 4.0      | 0                            | 0.0      |
| Total        | 87        | 68.0     | 22      | 17.0     | 19                           | 15.0     |

*B. The "Face-Up" Posture:* In 17 per cent of the fatalities, the "face-up" or supine posture was noted with the smothering agents completely covering the babies' faces. These agents were most commonly pillows and loosened blankets.

*C. Overlaying:* The next most frequent cause of smothering was by overlaying. This form of accidental death occurred in 15 per cent of the infants and mainly involved those from 2 weeks to 3 months of age. Smothering usually followed the removal of the infant from its crib to the mother's bed for the 2 A.M. or 6 A.M. feedings. Accidental death was brought about by occlusion of the infant's nose and mouth either by the breast or by one of the upper extremities of the mother who had fallen asleep. Occasionally, the infant was wedged in between both sleeping parents.

## COMMENT

To summarize, then, this study of accidental mechanical suffocation in infants under 1 year of age has disclosed the following circumstances underlying death:

1. The preponderance of deaths by smothering (85 per cent) occurred in infants under 6 months with the median age at death between 2 and 3 months.

2. Male babies showed a greater frequency of deaths (69 per cent) than female babies (31 per cent).

3. The peak of fatalities was reached during the late fall and early winter months, November through January.

4. When considered in relation to feeding hours, death from smothering took place most often between the hours from 3 A.M. to 6 A.M., and 6 A.M. to 9 A.M.

5. During the morning hours 35 per cent of the accidents occurred in cribs and 15 per cent in the mothers' beds. In the afternoon and night hours baby carriages and cribs were almost equally divided as the sites of accident.

6. In order of frequency the following were the smothering agents: (1) In the crib: blankets or outer coverings, mattresses, pillows, mattress coverings (sheets, pads and rubber sheetings); (2) in the carriage: pillows, blankets or outer coverings, mattresses and mattress coverings; (3) in the parent's bed: the mother's breast or upper extremity.

7. The posture most frequently noted on discovery of the infants' bodies was the "face-down" or prone posture (68 per cent). Fifty-five per cent of these infants were under 5 months of age.

8. In only 17 per cent of the fatalities were the bodies found in the "face-up" posture. Almost all these infants were 5 months of age and under.

9. In 46 per cent of the babies found in the "face-down" posture a single agent (pillow, mattress, or mattress covering) was in occluding contact with the noses and mouths, while in 22 per cent an added agent (pillow or blanket) covered the heads.

10. In the "face-up" posture pillows and blankets were found covering the infants' faces.

11. Death from overlaying comprised 15 per cent of the fatalities from smothering and involved very young infants from 2 weeks to 3 months of age. These accidental deaths occurred most frequently during the early morning hours.

## MEASURES OF PREVENTION

*Public Health Education.*—Naturally, the circumstances comprising the settings for accidental mechanical smothering in infancy immediately call to mind obvious remedial measures. The increasing occurrence of this form of accidental death in infancy should be more widely

publicized both in medical and in lay periodicals. Vigorous efforts should be made by public health agencies to acquaint parents, nurses, physicians, and others concerned with the care of infants with the hazards to which these babies are unwittingly exposed.

It is not generally appreciated that the chief cause of accidental death in early infancy is mechanical suffocation. This cause of death has reached a point where it exacts an annual toll of lives of infants under 1 year of age exceeding that from burns and other accidental fatalities. Lack of apprehension of this situation may be due in part to the infrequency with which the practicing physician and the hospital are called upon to attend these accidental cases of smothering. The investigation of such deaths falls within the province of the local police department and the office of the medical examiner. Stress is placed on determining the existence of suspicious violence. The data from these inquests are treated as strictly confidential and the details responsible for death are rarely reported in the medical literature.

*Care of the Infant.*—The extreme youth of the victims of accidental smothering emphasizes the constant attention which must be afforded these babies at all hours during the day and night. No item of care can be overlooked if deaths from suffocation are to be avoided. In the nursing supervision of the baby heed should be paid to the following details:

1. *Wearing Apparel:* (1) Sufficient clothing should be provided to maintain normal body temperature, but excessive bundling should be avoided. (2) Free and unhampered movement should be allowed the baby at all times. (3) Constrictions about the neck such as are caused by the tapes of caps or bibs or by tight collars of dresses and outer garments should be eliminated. (4) Sleeping bag garments with mechanical closing and locking devices which may catch directly beneath the infant's neck should be prohibited.

2. *The Crib and Carriage:* Greater attention should be paid to the preparation of the infant's crib and carriage. (1) No infant should ever sleep on a pillow, but always on a flat mattress. (2) Rubber sheets and undersheets should be spread smoothly and be sufficiently large to allow tight tucking beneath the mattress. (3) Quilted pads should be bound down or pinned down to prevent loosening. (4) Oversheets and blankets should be large enough to permit firm tucking beneath the sides and lower edge of the mattress, but at the same time not so tight as to prevent freedom of movement of the baby. (5) Unnecessary articles such as decorative pillows, odd unattached blankets, strings or bars from which toys are dangled, and restraining straps which may slip up about the neck should be eliminated.

3. *Feeding:* Considerable care should be exercised in the management of the infant during and after feeding hours when not only smothering but also choking can occur from aspiration of regurgitated food. (1) No practice should be more emphatically condemned than

that of placing for feeding the unattended infant in its mother's bed where death from overlaying may be caused by the mother who falls asleep. (2) Throughout feeding the infant should be held in the mother's or nurse's arm with the attendant properly seated. (3) In artificially fed infants the practice should likewise be condemned of feeding the baby unattended in its crib by propping up the bottle in the infant's mouth. (4) After feeding the infant should be held up for sufficient time to allow escape of the "air-bubble" and then be placed down gently on his back, not stomach, with the head temporarily elevated. (6) The frequent occurrence of smothering directly after feeding, particularly during the early morning hours, strongly indicates that the infant should not be left unguarded too soon after completion of a feeding.

4. *Position of Baby in Crib and Carriage:* One other factor of importance merits consideration with regard to the position the baby is put while in the crib or carriage. It is common nursing practice to place the infant in the "face-down" or prone position. This detail of nursing care has been carried out by mothers and nurses for the following reasons: Firstly, it is believed that the baby is more comfortable and falls asleep easily. Secondly, the procedure is recommended to prevent the flattening deformity of the skull or cranial asymmetry which results from continued rest in the recumbent posture.<sup>3</sup> This nursing procedure may be expedient during the hours when the infant is under constant watch. But consider the risks involved when the baby is left unguarded. Attention should again be called to the preponderance of infants under 5 months of age who were found in the "face-down" posture on death from smothering.

Explanation for the more frequent incidence of smothering among young infants who are placed in the prone position is probably found by studying their behavioral schedules with respect to motor development. According to Gesell<sup>4</sup> and others,<sup>5</sup> during the third week of life about 80 per cent of infants exhibit head turning movements from left to right when in the prone posture. Thereafter, this phase of cephalic motor activity shows a decrease in frequency and is superseded at about the ninth week by head and chin lifting off the table top or bed. At this time the duration of such movements is brief, lasting but a minute. It is not until the fourth to the sixth month that head and chin raising movements become established with, however, limited duration and strength of these activities. It should also be borne in mind that the appearance of body turning or rolling movements have not begun.

It is, therefore, not surprising that of the infants under 6 months of age who died from accidental smothering so many were found in the "face-down" position in which they were previously placed. Head-and chin-lifting motor activities are not sufficiently developed to overcome the suffocating weight of the pillow or the blanket over the baby's head; nor are they of sufficient duration and strength to prevent con-

tinued occluding contact of the nose and mouth with the surface of the mattress to the point of smothering. It seems reasonable, therefore, to recommend that the routine nursing practice of placing infants in the prone position be avoided except during such times as the babies are constantly attended. The practice should, furthermore, be entirely done away with at night.

#### SUMMARY

Attention is called to the excessive number of infants under 1 year of age who are dying from accidental mechanical suffocation. These fatalities have shown a progressive increase each year to the point where accidental mechanical smothering now holds the dubious distinction of being the leading cause of accidental death in early infancy. Yet astonishingly little concern has been evinced with regard to the problem by public health workers and the literature contains but few comments. Concerted efforts should be made to study the problem completely and a constructive program should be designed and disseminated to prevent deaths from accidental mechanical suffocation.

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